Technical Blueprint

SAFESPHERE

AI + Statistical Disaster Intelligence Platform

- Hackathon-Ready Architecture
- Lightweight Statistical Models
- Single Server Deployment

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Version 1.0

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Complete technical stack from frontend to database

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Supervised, unsupervised AI & statistical modeling

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Historical disaster intelligence & anomaly detection

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Forecasting risk, impact & resource demand

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Technical Foundation

System Architecture

Frontend Layer

React
Component-based UI framework

Chart.js
Interactive data visualization

Leaflet
Open-source mapping library

Service Worker
Offline functionality support

AI + Statistical Engine

- Logistic Regression
- Linear Regression
- K-Means Clustering
- Z-Score Analysis
- Statistical Modeling

Backend Layer

Node.js
JavaScript runtime environment

Express
RESTful API framework

Data Layer

MySQL Database
Relational data storage with optimized schema

Single Server Deployment | Lightweight Models | Hackathon Ready | REST API

AI & Analytics

Intelligence Stack

Supervised AI
Predictive Models

Logistic Regression
Binary classification algorithms

Disaster Risk Prediction | Emergency Priority

Linear Regression
Continuous value forecasting

Impact Forecast | Resource Demand

Unsupervised AI
Pattern Discovery

K-Means Clustering
Data segmentation algorithm

Region Vulnerability | Grouping Analysis

Anomaly Detection
Z-score spike detection $|Z| > 2$

Statistical Modeling
Data Processing

- Z-score normalization
- Growth rate analysis
- Moving average
- Variance & standard deviation
- Percentile classification
- Correlation analysis
- Confidence intervals

NLP Removed: Lightweight statistical approach without natural language processing

Hackathon-Optimized

Historical Intelligence

Past Analyzer Page

Trend Analysis Features

5-Year Frequency
Disaster occurrence patterns

Severity Trend
Impact intensity tracking

Economic Loss
Financial impact analysis

Moving Average
Trend smoothing

Statistical Metrics

Growth Rate
Year-over-year change

Mean & Std Dev
Central tendency & spread

Variance Analysis
Data dispersion measure

Correlation
Risk vs loss relationship

Anomaly Detection

$$Z = (X - \mu) / \sigma$$

X	Observed value
μ	Mean
σ	Standard deviation

Alert Trigger
If $|Z| > 2 \rightarrow$ **Anomaly Alert**

Clustering

K-Means Algorithm
Groups regions by vulnerability characteristics

- High-risk clusters
- Medium-risk clusters
- Low-risk clusters

Purpose: Historical disaster intelligence for pattern recognition and baseline establishment

Predictive Analytics

Future Prediction Page

A. Disaster Forecast

$$y = mx + c$$

Linear Regression Model

Predicts: Frequency, Severity

Confidence Interval

CI = Prediction ± 1.96 × SE

B. Risk Forecast

Logistic Regression

Converts predicted metrics → probability

75%

High-risk probability

C. Impact Forecast

- Affected Population
- Economic Loss
- Infrastructure

D. Resource Demand

$$Demand = \alpha + \beta_1(Risk)$$

Linear Regression Model

Shortage Alert

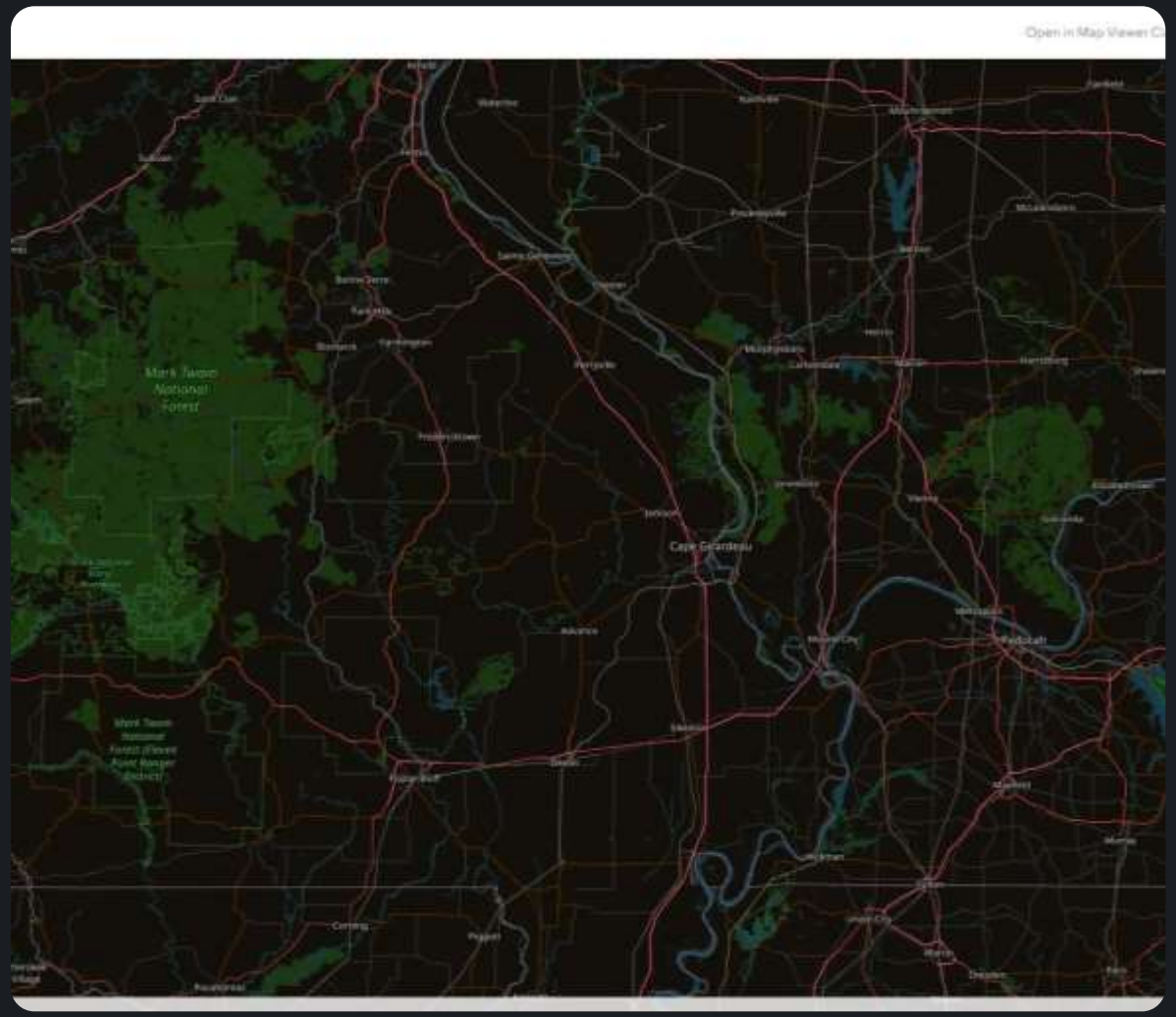
If Stock < Demand + 15%

Future Anomaly Warning: If predicted value > mean + 2σ → warning triggered

Proactive Intelligence

Geospatial Analytics

GIS Intelligence Page



Core Features

Percentile-Based Color Shading
Risk levels visualized through color gradients

Disaster Type Toggle
Filter by flood, earthquake, fire, etc.

Interactive Analytics
Hover breakdown & click-to-region

Marker Systems

Emergency Markers
Active incidents & response units

Shelter Markers
Safe zones & capacity status

Anomaly Overlay
Z-score detected hotspots highlighted on map

Technology: Leaflet.js with custom tile layers, real-time data binding, and interactive event handlers

Response Coordination

Emergency Management Page

Emergency Inputs

- Name
- Location
- Disaster Type
- Medical Severity (1-5)
- Age Group
- Risk Zone (Auto-filled)

Workflow Stages

1. Pending →
2. Assigned →
3. In Progress →
4. Resolved

AI Priority Model

Priority = (0.45 × Medical) + (0.25 × Age) + (0.20 × Risk) + (0.10 × Density)

Medical Severity	45%
Age Vulnerability	25%
Risk Zone Level	20%
Population Density	10%

Logistic Regression

Predicts probability of high urgency classification

Management Features

- Admin Assignment
- Response Tracking
- Priority Sorting

Resource Optimization

Shelter & Resource Management

Shelter Intelligence

Add/Edit Shelters
Dynamic shelter registry

Occupancy Tracking
Real-time capacity %

Overcrowding Alert
If Occupancy > mean + 1.5σ ⟶ warning

Future Forecast
Future = Current + (GrowthRate × Days)

Alert Systems

Shortage Alert
If Stock < Demand + 15%

Safety buffer maintained

Correlation Insights

Risk vs Water Demand	r = 0.82
Risk vs Food Demand	r = 0.76

Resource Management

Add/Update Stock
Inventory management

Demand Prediction
Linear regression forecast

Resource Categories

- Water
- Food
- Medical
- Bedding

Offline Resilience

Emergency Offline Page

Technology Stack

Service Worker
Background script for offline functionality and network interception

IndexedDB
Client-side storage for emergency data persistence

Background Sync
Automatic data synchronization when connection restored

Processing Logic

Local Processing
Basic statistical scoring

Synced Processing
Full ML classification

Offline Workflow

1. **Submit Emergency**
   User fills emergency form
2. **Save Locally**
   Data stored in IndexedDB
3. **Show "Saved Offline"**
   User confirmation message
4. **Auto-Sync on Reconnect**
   Background Sync triggers

Zero Data Loss: All emergencies captured regardless of connectivity

Data Architecture

Database Structure & System Flow

Database Schema

regions
id, name, population, climate_factor

disaster_history
region_id, year, frequency, severity, loss

risk_predictions
risk_index, probability, std_dev, anomaly

emergency_requests
priority, probability, status

shelters
capacity, occupancy

resources
quantity, threshold

clusters
region_id, cluster_group

users
role, credentials

MySQL Database

Relational data storage with optimized schema for disaster intelligence operations

Core System Flows

Risk Engine Flow

Historical Data → Normalize → Growth → Logistic → Percentile → Anomaly → Heatmap

Emergency Flow

Input → Statistical → Logistic → Priority → Workflow

Resource Flow

Risk → Impact → Demand → Buffer → Alert

Architecture: Modular, scalable, and optimized for real-time disaster response operations

SafeSphere

Project Identity

SafeSphere is a **statistically grounded, AI-powered disaster intelligence platform** that predicts risk, detects anomalies, forecasts impact, prioritizes emergencies, and manages shelters and resources — **without heavy AI complexity**.

AI-Powered	Lightweight	Comprehensive
Statistical models	Hackathon ready	End-to-end platform

Final Feature Checklist

- Multi-hazard risk prediction
- Linear regression forecasting
- K-means clustering
- Past analyzer
- GIS heatmap
- Shelter analytics
- Correlation analysis
- Role-based access
- Logistic regression modeling
- Z-score anomaly detection
- Confidence intervals
- Future prediction
- Emergency ML classification
- Resource demand prediction
- Offline emergency support
- Reports export

Platform Strengths

Lightweight
No heavy AI frameworks, statistical models only

Single Server
Simplified deployment architecture

Hackathon Ready
Rapid development & demo capability

Offline Capable
Service Worker & IndexedDB support

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